

Mandatory Assignment 2

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Exercise 1.

1a: Variable selection and sample restriction

First, we remove observations before year 2000, then we select ten stock-level characteristics based on the variable-importance results in [Gu et al. \(2020\)](#), Figure 5. The selected variables cover price trends: `mom1m`, `mom12m`, size and liquidity: `mvel1`, `turn`, risk and volatility: `beta` `retvol`, and valuation/fundamentals: `bm`, `ep`, `roaq`, `invest`.

For the macroeconomic state variables from [Goyal and Welch \(2008\)](#) library, we follow [Gu et al. \(2020\)](#) Table 4 and include aggregate book-to-market (`bm`), the Treasury bill rate (`tbl`), the default yield spread (`dfy`), and the term spread (`tms`). These variables capture aggregate valuation, interest rate conditions, credit risk, and the slope of the yield curve.

1b: Summary Statistics and Missing Values

Table 1 reports summary statistics for the selected predictors. Stock characteristics are approximately centered around zero and bounded close to $[-1, 1]$, while macro variables vary at the monthly aggregate level.

	mean	std	min	25%	50%	75%	max
<code>characteristic_mom1m</code>	-0.004	0.604	-0.996	-0.550	-0.009	0.544	0.995
<code>characteristic_mom12m</code>	-0.007	0.590	-0.997	-0.526	-0.026	0.519	0.992
<code>characteristic_mvel1</code>	0.026	0.588	-1.000	-0.496	0.044	0.544	1.000
<code>characteristic_turn</code>	0.036	0.576	-1.000	-0.453	0.074	0.532	0.994
<code>characteristic_beta</code>	0.076	0.548	-0.993	-0.370	0.102	0.540	0.993
<code>characteristic_retvol</code>	0.123	0.528	-1.000	-0.306	0.146	0.576	0.996
<code>characteristic_bm</code>	-0.065	0.531	-0.990	-0.512	-0.067	0.373	0.990
<code>characteristic_ep</code>	-0.011	0.558	-0.990	-0.486	-0.004	0.454	0.990
<code>characteristic_roaq</code>	-0.014	0.560	-0.991	-0.477	-0.023	0.457	0.990
<code>characteristic_invest</code>	-0.006	0.533	-0.990	-0.427	-0.000	0.413	0.990
<code>macro_bm</code>	0.276	0.072	0.121	0.239	0.284	0.328	0.446
<code>macro_tbl</code>	0.018	0.019	0.000	0.001	0.011	0.028	0.062

	mean	std	min	25%	50%	75%	max
macro_dfy	0.010	0.004	0.005	0.008	0.009	0.012	0.034
macro_tms	0.022	0.014	-0.004	0.009	0.023	0.035	0.045

We next document missing values before applying the imputation procedure.

	n_missing
characteristic_mom1m	0
characteristic_mom12m	0
characteristic_mvell	0
characteristic_turn	0
characteristic_beta	0
characteristic_retvoll	0
characteristic_bm	0
characteristic_ep	0
characteristic_roaq	0
characteristic_invest	0
macro_bm	0
macro_tbl	0
macro_dfy	0
macro_tms	0

Missing stock characteristics are imputed using the monthly cross-sectional median. This preserves the number of stock-month observations while using only contemporaneous information from the same month. The median is preferred to the mean because it is less sensitive to extreme firm-level observations, which are common in financial data.

Macroeconomic variables are common to all firms in a given month and vary only over time. Missing macro values are therefore filled forward and backward to preserve the time series of aggregate state variables. Finally, observations with missing excess returns or lagged market capitalization are dropped because `ret_excess` is the prediction target and `mktcap_lag` is required to construct value-weighted portfolio returns.

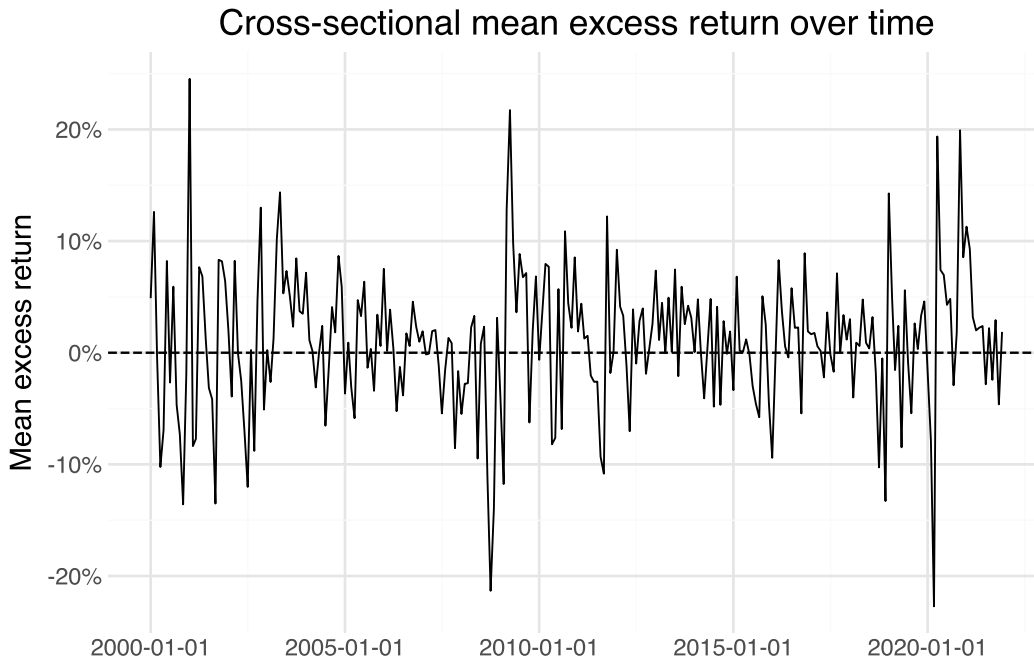
	n_missing
characteristic_mom1m	0
characteristic_mom12m	0
characteristic_mvell	0
characteristic_turn	0
characteristic_beta	0
characteristic_retvoll	0
characteristic_bm	0
characteristic_ep	0
characteristic_roaq	0
characteristic_invest	0

	n_missing
macro_bm	0
macro_tbl	0
macro_dfy	0
macro_tms	0
ret_excess	0
mktcap_lag	0

We recode `sic2` into the 10 major SIC divisions to retain broad industry information while keeping the feature space manageable. This reduces the number of industry dummy variables and lowers the risk that sparse industry categories create noisy predictors in the machine learning models.

	sic_division	n_obs
0	Manufacturing	442708
1	Finance	224201
2	Services	194522
3	Transportation	86802
4	Retail	64355
5	Mining	42998
6	Wholesale	31682
7	Construction	12262
8	Public Admin	9530
9	Agriculture	2545

Figure 1 plots the monthly cross-sectional mean excess return. The series illustrates that average realized excess returns vary substantially over time, highlighting the noisy nature of stock return prediction.



Overall, the sample remains large after cleaning and is suitable for machine learning. The missing-value treatment balances data retention with robustness, while the industry recoding keeps the dimensionality of the feature matrix manageable.

1c: Feature matrix construction

We construct the feature matrix using the selected stock characteristics and their interactions with macroeconomic state variables. The feature matrix contains 59 predictors: 10 stock characteristics, 40 characteristic-macro interactions, and 9 industry dummies. The interaction terms allow firm-level return predictability to vary with macroeconomic conditions, consistent with [Gu et al. \(2020\)](#). One SIC division is omitted as the reference category when constructing industry dummies.

Exercise 2

Exercise 2a: Model descriptions

Elastic Net is a penalized linear regression model that combines the variable selection property of the Lasso with the coefficient shrinkage of Ridge regression. It is useful in our setting because the feature matrix contains many correlated predictors, including characteristic-macro interactions. The model minimizes prediction error while penalizing large coefficients, which helps reduce overfitting.

The key hyperparameters are the overall penalty strength (**alpha**) and the mixing parameter (**l1_ratio**). A higher **alpha** implies stronger regularization, while **l1_ratio** controls the balance between Lasso and Ridge. When **l1_ratio** = 1, the model corresponds to the Lasso; when **l1_ratio** = 0, it corresponds to Ridge regression.

Random Forest is a nonlinear tree-based ensemble model. It builds many regression trees on different bootstrap samples of the data and averages their predictions. Each tree recursively splits the feature space into groups with similar realized excess returns. Averaging across trees reduces the instability of individual decision trees and helps improve out-of-sample performance.

The key hyperparameters are the number of trees (**n_estimators**), the number of predictors considered at each split (**max_features**), the minimum number of observations required in a leaf (**min_samples_leaf**), and tree depth (**max_depth**). Compared with Elastic Net, Random Forest is more flexible because it can capture nonlinearities and interactions without imposing a linear functional form.

Together, the two models provide a comparison between a regularized linear benchmark and a flexible nonlinear machine learning model.

Exercise 2b: Time-respecting data split

	Sample	Start date	End date	Months	Observations
0	Train	2000-01-01	2013-02-01	158	748902
1	Validation	2013-03-01	2017-07-01	53	194460

	Sample	Start date	End date	Months	Observations
2	Test	2017-08-01	2021-12-01	53	168243

The dataset is split chronologically to avoid look-ahead bias. Training observations precede validation observations, which in turn precede test observations. The most recent 20% of months are reserved for the test set and are only used for final out-of-sample evaluation.

The remaining 80% of months is split into training and validation samples, yielding an approximate 60/20/20 split. The training set is used to estimate model parameters, while the validation set is used for hyperparameter tuning. The split is performed by month rather than by individual rows to preserve the panel structure.

Continuous predictors are standardized using the training sample only. The same transformation is then applied to the validation and test sets, which avoids data leakage from future observations.

Exercise 2c: Hyperparameter tuning

	alpha	l1_ratio	train_rmse	val_rmse
6	0.100	0.2	0.199689	0.157523
7	0.100	0.5	0.199689	0.157523
5	0.010	0.8	0.199689	0.157523
8	0.100	0.8	0.199689	0.157523
4	0.010	0.5	0.199682	0.157524
3	0.010	0.2	0.199484	0.157563
0	0.001	0.2	0.199279	0.157587
2	0.001	0.8	0.199378	0.157640
1	0.001	0.5	0.199346	0.157652

	n_estimators	max_features	min_samples_leaf	max_depth	train_rmse	val_rmse
1	30	sqrt	100	10	0.195209	0.157395
0	30	sqrt	100	5	0.197974	0.157426

	Model	Train RMSE	Validation RMSE
0	Elastic Net	0.199689	0.157523
1	Random Forest	0.195209	0.157395

	Model	alpha	l1_ratio	n_estimators	max_features	min_samples_leaf	max_depth
0	Elastic Net	0.1	0.2	NaN	NaN	NaN	NaN
1	Random Forest	NaN	NaN	30.0	sqrt	100.0	10.0

Hyperparameters are tuned using grid search. Each candidate specification is fitted on the training sample and evaluated on the validation sample using RMSE, consistent with the mean squared prediction error objective in [Gu et al. \(2020\)](#). For Elastic Net, the grid varies the overall penalty strength (`alpha`) and the balance between Lasso and Ridge regularization (`l1_ratio`). For Random Forest, the grid varies the number of predictors considered at each split (`max_features`), the minimum leaf size (`min_samples_leaf`), and the maximum tree depth (`max_depth`).

The final specification for each model is selected as the one with the lowest validation RMSE. Comparing training and validation RMSE illustrates the bias-variance trade-off: more flexible models may fit the training data better, but this does not necessarily translate into better validation performance.

Exercise 2d: Save models to disk

```
['model_random_forest.joblib']
```

To load the models for peer review, run:

```
final_en = joblib.load("model_elastic_net.joblib")
final_rf = joblib.load("model_random_forest.joblib")
```

The final fitted models are saved to disk using `joblib` to ensure reproducibility and facilitate peer review. Saving the trained models allows others to load the exact same fitted objects without re-running the full hyperparameter tuning procedure.

Exercise 3

Exercise 3a: Generate predicted excess returns

Using the fitted models, we generate predicted excess returns for each stock-month observation in the validation and test samples. These predictions reflect expected returns based on firm characteristics, macroeconomic variables, and their interactions. The predictions are used to rank stocks in portfolio construction, where higher predicted returns correspond to higher deciles and more attractive investment opportunities.

Exercise 3b: Decile portfolio sorts

Stocks are sorted into decile portfolios each month based on predicted excess returns. Within each month, stocks are ranked and divided into ten groups of approximately equal size, from the lowest (decile 0) to the highest (decile 9).

To ensure stable portfolio formation, predictions are ranked before applying quantile sorting, avoiding issues with tied values. Months with very few observations are excluded.

This procedure generates a cross-sectional ranking that reflects the model's expected return signal and is used to construct long-short portfolios.

Exercise 3c: Long-short portfolio

Exercise 3d: Performance metrics

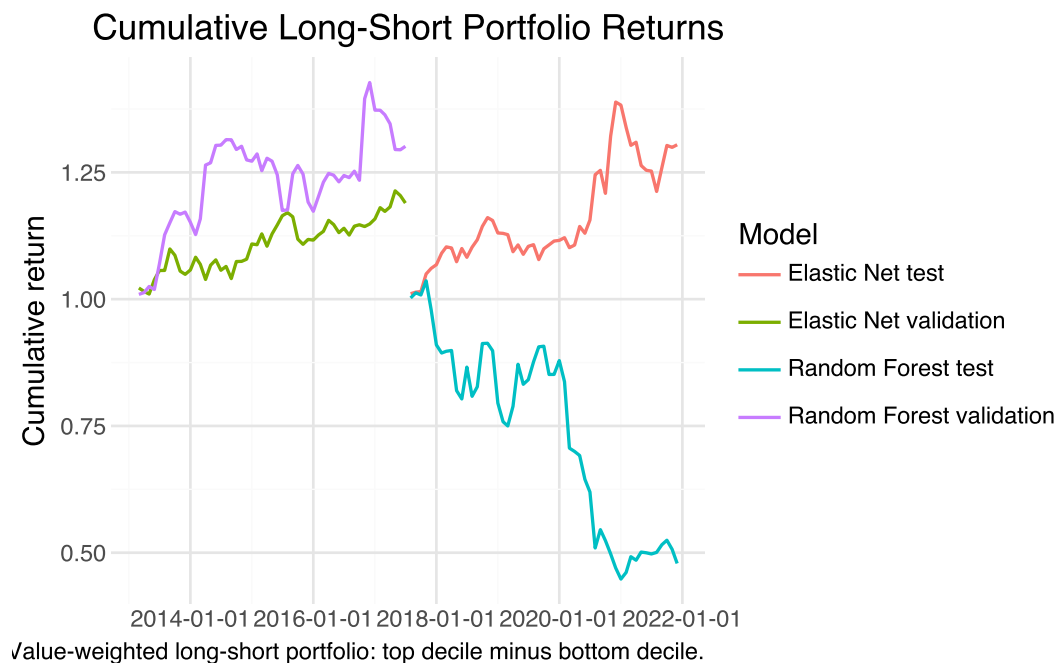
The market benchmark is constructed as the value-weighted average excess return across all stocks in the relevant sample. This makes the benchmark comparable to the long-short portfolio, since both returns are measured in excess of the risk-free rate. CAPM alpha is estimated by regressing the long-short portfolio excess return on the value-weighted market excess return.

	Model	Sample	ann_mean	sharpe	alpha	alpha_t
0	Elastic Net	Validation	0.0410	0.6976	0.0405	1.3369
1	Random Forest	Validation	0.0653	0.6084	0.0344	0.6369
2	Elastic Net	Test	0.0641	0.7228	0.0449	1.0329
3	Random Forest	Test	-0.1466	-0.7588	-0.1069	-1.1267

Exercise 3e: Portfolio turnover

	Model	Validation	Test
0	Elastic Net	0.007	0.008
1	Random Forest	0.559	0.544

Exercise 3f: Model comparison



Both models are compared across Sharpe ratio, CAPM alpha, and portfolio turnover. Elastic Net is the regularized linear benchmark and tends to produce smoother predictions because it imposes

a linear structure and coefficient shrinkage. Random Forest is more flexible and can capture nonlinearities and interactions between predictors, but this flexibility may also lead to higher portfolio turnover if the ranking of stocks changes more frequently over time. From an investor's perspective, the preferred model is the one that delivers stronger risk-adjusted performance and alpha without requiring excessive turnover.

Exercise 4

Exercise 4a: Variable importance

	model	feature	sharpe_drop
0	Elastic Net	characteristic_mom1m	0.0000
1	Elastic Net	characteristic_mom12m	0.0000
2	Elastic Net	characteristic_mvell	0.0000
3	Random Forest	characteristic_beta	0.4016
4	Random Forest	characteristic_mom1m_x_macro_tms	0.2046
5	Random Forest	characteristic_beta_x_macro_dfy	0.1751

Variable importance is measured as the drop in annualized test Sharpe ratio when one predictor is set to a reference value and the long-short portfolio is rebuilt. For standardized continuous predictors, the reference value is zero, corresponding to the training-sample mean after standardization. For industry dummies, the reference value is set to the test-sample mean.

This portfolio-based importance measure evaluates whether a predictor matters for the final investment strategy rather than only for return prediction accuracy. A larger Sharpe-ratio drop indicates that removing the information in that predictor reduces the economic value of the model.

Exercise 4b: Ranking comparison

	feature	rank_en	rank_rf	rank_gap
7	characteristic_bm_x_macro_dfy	1.0	59.0	58.0
41	characteristic_roaq_x_macro_bm	1.0	58.0	57.0
5	characteristic_bm	1.0	57.0	56.0
12	characteristic_ep_x_macro_dfy	1.0	56.0	55.0
39	characteristic_retvoll_x_macro_tms	1.0	55.0	54.0

The rankings differ across the two models because Elastic Net and Random Forest use the predictors in different ways. Elastic Net is a regularized linear model, so predictor importance is tied to linear effects after shrinkage. Random Forest is more flexible and can assign importance to predictors that matter through nonlinear patterns or interactions. Differences in the rankings therefore reflect differences in model structure, not only differences in predictive strength.

Exercise 4c: Findings to financial theory

The variable-importance results can be interpreted in relation to financial theory. If momentum-related variables rank highly, this is consistent with return continuation or reversal effects. If valuation variables such as book-to-market or earnings-to-price are important, the results point toward value-based predictability. Risk-related variables such as beta or return volatility can be interpreted as compensation for systematic or idiosyncratic risk, while industry dummies suggest that expected returns vary across sectors.

Because Random Forest can capture nonlinearities and interactions, its most important variables may reflect state-dependent effects rather than simple standalone return premia. Overall, the variable-importance exercise links the machine-learning predictions back to economically interpretable return predictors.

References

- Goyal, A. and Welch, I. (2008). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.